Architecture 1001:
x86-64 Assembly
First Assembly Instruction:
NOP

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Your first x86-64 instruction: NOP

- NOP - No-Operation! No registers, no values, no nothin'!
- Just there to pad/align bytes, or to delay time
- Attackers use it to make simple exploits more reliable. But that's another class ;)

Extra! Extra! Late-breaking NOP news!

- Amaze those who know x86 by citing this interesting bit of trivia:
- “The one-byte NOP instruction is an alias mnemonic for the XCHG (E)AX, (E)AX instruction.”
  - I had never looked in the manual for NOP apparently :)
- XCHG instruction is not officially in this class. But if I hadn’t just told you what it does, I bet you would have guessed right anyway

0x90

Instructions we now know (1)

- NOP

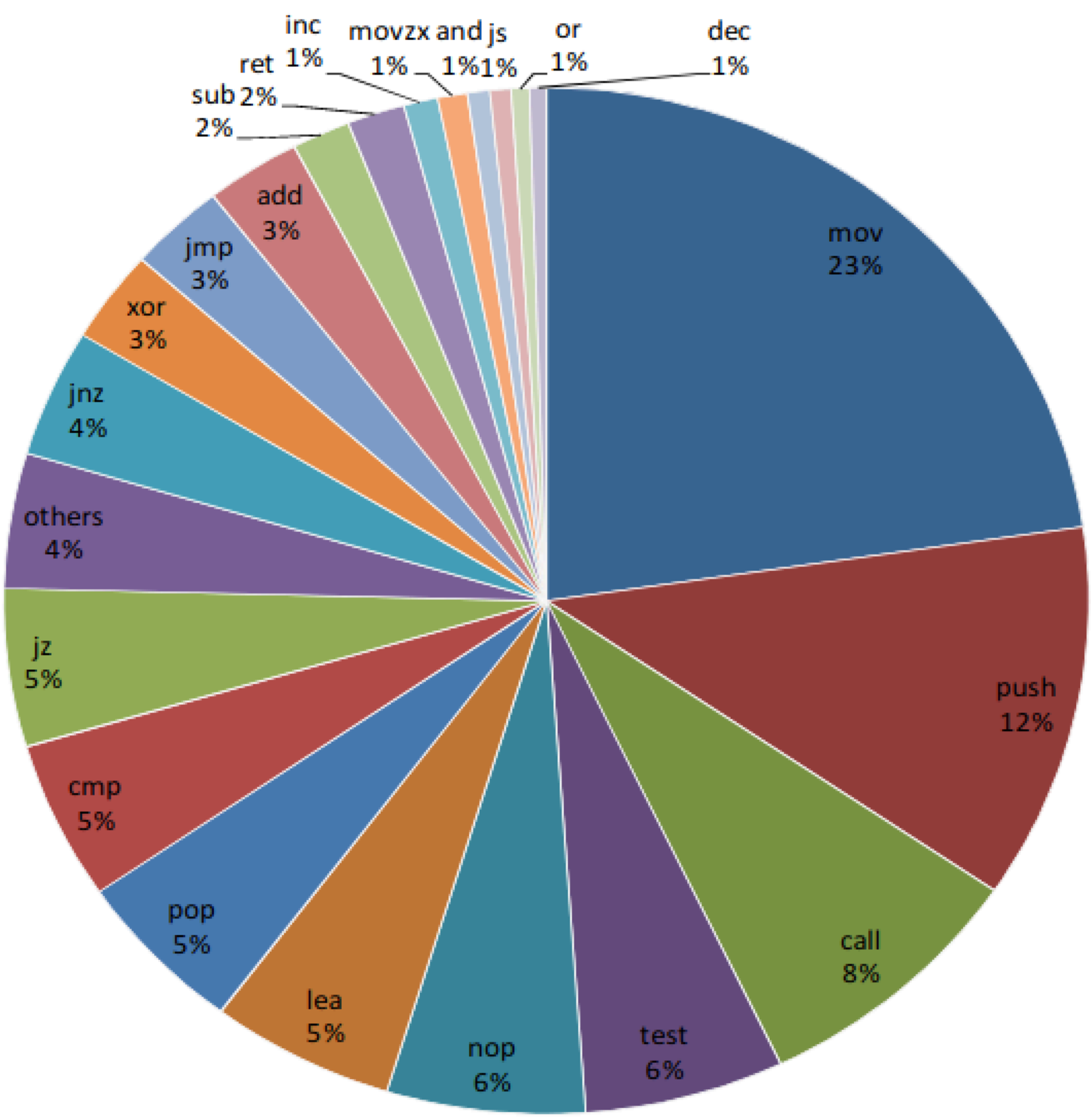


Fig. 2 Web browsers' instruction frequencies

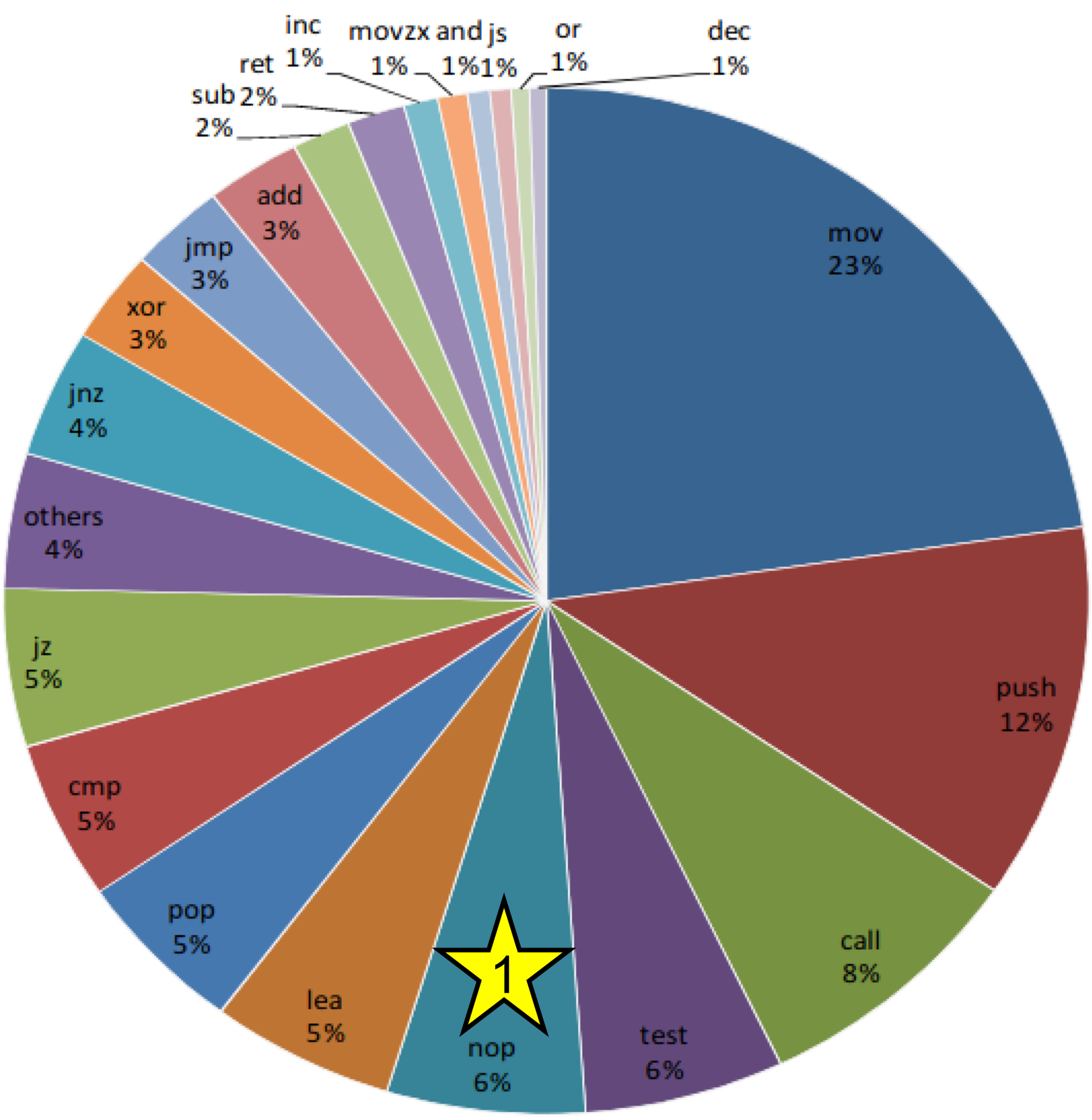


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